

OPTIMISING VOCABULARY RETENTION THROUGH SPACED REPETITION

A Data-Driven Analysis of Duolingo Word Recall Patterns

Word Recall Rate | Lag & Exposure Analysis | Course & Language Comparison

90.4%

OVERALL WORD RECALL RATE

908,795 practice attempts · 499,436 session records · 75,294 learners · 11,227 words · 6 target languages

3.8 pts	Flat	1.3 pts	9% / 7.3 pts
Recall drop, <1 hour to 3+ months since last practice	Recall vs. prior exposure count	Function-word vs. content-word gap	of volume, hardest 20% of words, recall gap

Next Leap Data Analyst Fellowship — Graduation Project

Dataset provided by Mind Lab — a sample of the public Duolingo Half-Life Regression release *NL_Duolingo_SRS*

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How This Report Is Built

Metric definition: every recall percentage in this report is ***attempt-weighted*** — SUM(session_correct) ÷ SUM(session_seen) — never a simple average of per-group rates. This matters because an unweighted average would let a low-volume language or course count as much as a high-volume one, quietly distorting every comparison.

Colour language: charts throughout this report use one consistent scale — **green** marks strong recall, **amber** marks the mid-range, and **coral / red** marks the weakest recall in that chart's own range — so colour always means the same thing, chart to chart.

Structure: Sections 1–4 set up the business question and the data; Section 5 shows the two patterns that shape everything after it; Section 6 tests five specific hypotheses, each with its own chart, statistical test, and recommendation; Sections 7–9 close with prioritised actions, limitations, and sources.

1. Executive Summary

Duolingo's spaced-repetition engine decides when to re-show a previously learned word so it is neither forgotten nor over-practiced. This analysis uses a 12-day sample of practice-session data (**499,436 session records, 908,795 individual practice attempts, 75,294 learners, and 11,227 unique words across 6 target languages**) to identify where the current, largely uniform scheduling approach is leaving recall on the table.

3.8 pts Recall drop from <1 hour to 3+ months since last practice	**Flat** Recall vs. prior exposure count — repetition alone doesn't help	**1.3 pts** Function-word vs. content-word recall gap	**9% / 7.3 pts** of volume comes from the hardest 20% of words, recalling 7.3 pts worse

Top Findings

- Recall decays measurably with time since last practice (**91.7% within 1 hour vs. 87.9% after 3+ months**) — but this pattern is invisible to a naive row-level correlation (Spearman $\rho \approx -0.047$) and only appears once lag is analysed in bins with attempt-weighted aggregation (total correct ÷ total attempts, not a simple average of averages).
- Prior exposure count alone is a **flat predictor** of recall (90.2%–90.8% across all five exposure bins, no clear trend). Timing, not repetition volume, is the lever that matters.
- **Function words** (determiners, prepositions, pronouns, conjunctions) recall at **89.5%**, versus **90.8%** for content words (nouns, verbs, adjectives) — consistent with known second-language-acquisition patterns.
- Course pairing affects recall independent of target language alone — **en** → **fr** shows the lowest recall (**89.4%**) of 8 course pairings; **it** → **en** the highest (**91.4%**).
- The hardest-ranked 20% of words account for roughly **9%** of all practice volume yet recall **7.3 points lower** on average (**83.8% vs. 91.1%**) than the remaining 80% of the vocabulary — a concentrated, targetable problem.

Headline Recommendation

Move from a uniform review interval to a scheduling model that weights **lag recency, word class (function vs. content), and word-level difficulty rank** — exposure count alone is not a reliable lever, but these three factors are, and they are all measurable in the data Duolingo already collects.

2. Business Context & Problem Framing

2.1 About Duolingo

Duolingo is a language-learning platform serving hundreds of millions of learners across dozens of languages through short, gamified daily practice sessions. A core part of the product experience is spaced repetition: deciding when to re-show a previously learned word so that it is neither forgotten nor over-practiced.

2.2 Business Problem

As the platform scales its course catalog and learner base, review scheduling is applied largely uniformly rather than tuned per language, per course, or per word. This analysis identifies where that uniform approach breaks down, using empirical recall patterns rather than assumptions about learner behaviour.

2.3 Objective

The goal is not simply to maximise the observed recall rate — a scheduler that only ever re-shows easy, already-mastered words would post a near-100% recall rate while teaching nothing new. The real objective is calibrating review timing so recall sits in a ***healthy target band***: reviews shown neither so late that the word is forgotten (frustration, session abandonment) nor so early that the review feels redundant (wasted practice slot, slower progression).

2.4 Business Impact

Word Recall Rate functions as a leading-indicator proxy for downstream engagement metrics that this dataset does not directly contain — session completion, streak retention, and long-term course completion. Better-calibrated scheduling is expected to improve perceived learning efficacy per session, which is the mechanism by which spaced repetition is believed to support retention and, ultimately, subscription conversion.

> *Word Recall Rate is used throughout this report as: SUM(session_correct) ÷ SUM(session_seen) — an attempt-weighted ratio, not an average of per-group rates. This is the same definition applied consistently across every chart and table below.*

3. Dataset Overview

The dataset is split across four related files rather than one flat table. learning_traces_sample.csv is the only row-level fact table (one row = one word practiced in one session); the other three are pre-aggregated rollups of it, used for cross-validation and fast summary views.

File	Grain	Rows	Role
learning_traces_sample.csv	session × word	499,436	Row-level source of truth
word_difficulty.csv	word (lexeme)	6,211	Word-level difficulty rollup
language_courses.csv	course pairing	8	Course-level rollup
forgetting_curve.csv	lag_bin × practice_bin	311	Pre-aggregated decay curve

Join keys: lexeme_id links traces to word_difficulty; ui_language + learning_language (combined into a course label) links traces to language_courses. forgetting_curve is a macro validation layer, already binned, and joins on learning_language + lag_bin + practice_bin only.

3.1 Key Metric Tree

Word Recall Rate (session_correct ÷ session_seen, attempt-weighted) is the top-level metric. It is driven by four groups of factors: *time-decay* (lag_days / lag_bin), *practice strength*

(history_seen / prior exposures), *content* (part of speech, word-level difficulty rank), and *structural factors* (target language, course pairing).

3.2 Coverage & Scale

- 908,795 practice attempts across 499,436 session-word records, spanning Feb 28 – Mar 12, 2013 (a 12-day window, not a longitudinal log)
- 75,294 unique learners; 11,227 unique words (lexemes)
- 6 target (learning) languages: English, Spanish, French, German, Italian, Portuguese
- 4 interface (native) languages, forming 8 distinct course pairings — volume is heavily concentrated in two of them

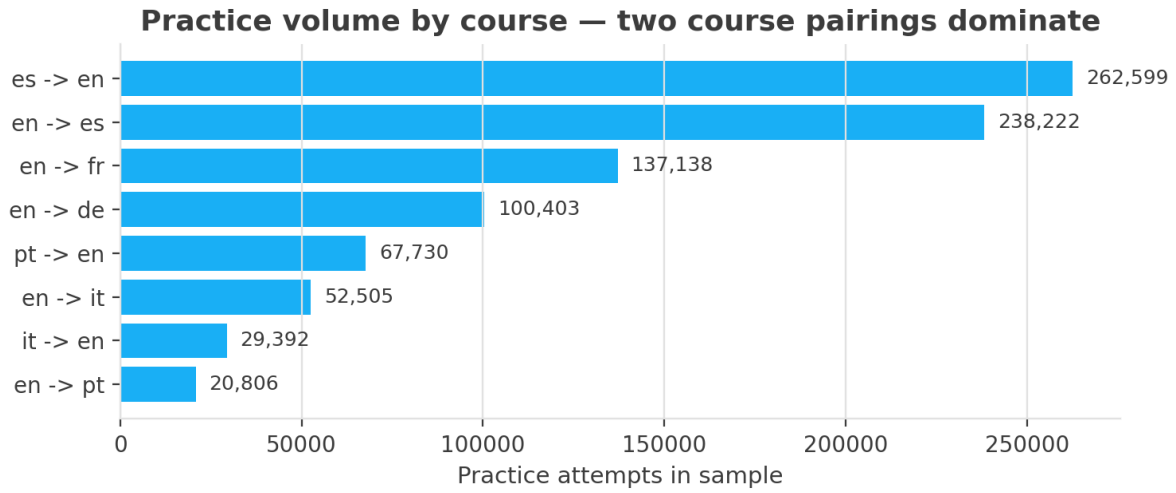


Figure 0. es → en and en → es alone account for well over half of all practice attempts in the sample — any scheduling change tested here reaches statistical significance fastest on these two courses.

4. Data Cleaning & Preparation

Only decisions that materially affected the analysis are documented below; routine checks that found nothing (duplicate rows: zero; dtype issues: none) needed no corrective action.

4.1 grammar_tags nulls (≈ 10.2% of records)

These are not missing measurements — some parts of speech (certain prepositions, conjunctions) genuinely carry no grammar tag. Nulls were filled with an explicit 'none' category rather than imputed, so the field remains a valid, non-misleading category in any group-by.

4.2 Outliers in lag_days and history_seen

Both fields are heavily right-skewed: *history_seen* has a median of 6 but a maximum of 13,447; *lag_days* has a median of 0.9 days but a maximum of 466 days. Raw values were not deleted or capped (they are valid observations, likely reflecting genuine power users) — instead, both fields were **binned** for any analysis where a raw mean would let a handful of extreme cases dominate the picture. Bin edges were set to match forgetting_curve.csv's own categories exactly, so hand-computed and pre-aggregated figures remain cross-checkable against each other.

4.3 Word-Class Derivation

A binary content word vs. function word flag was derived from the part-of-speech field (determiners, prepositions, pronouns, and conjunctions classified as function words; all others as content words), based on established second-language-acquisition distinctions between grammatically-structural and semantically-independent vocabulary.

4.4 Merge Integrity

Before merging traces with word_difficulty on lexeme_id, uniqueness of lexeme_id in word_difficulty was verified (confirmed one row per word) to guarantee the merge does not silently duplicate rows.

Baseline established: the overall, attempt-weighted Word Recall Rate is 90.4% (822,036 correct of 908,795 attempts). Roughly 1 in 10 recall attempts fails. Every section below decomposes that number to find where the failures concentrate.

5. Exploratory Data Analysis — Headline Findings

Two patterns emerged during exploration that shape how every hypothesis below is framed.

5.1 The Raw-Correlation vs. Binned-Aggregate Contradiction

At the row level, *lag_days* and *history_seen* show near-zero linear correlation with recall (Spearman ρ = -0.047 and ρ = +0.002 respectively). Yet the pre-aggregated, binned view below shows a clean, textbook forgetting curve. This is **not a data error** — it is confounding: a hard word reviewed early can score worse than an easy word reviewed late, washing out the raw correlation. *No factor in this dataset can be treated as a lone predictor* — each needs to be viewed through bins or in combination with other factors, not raw pairwise correlation.

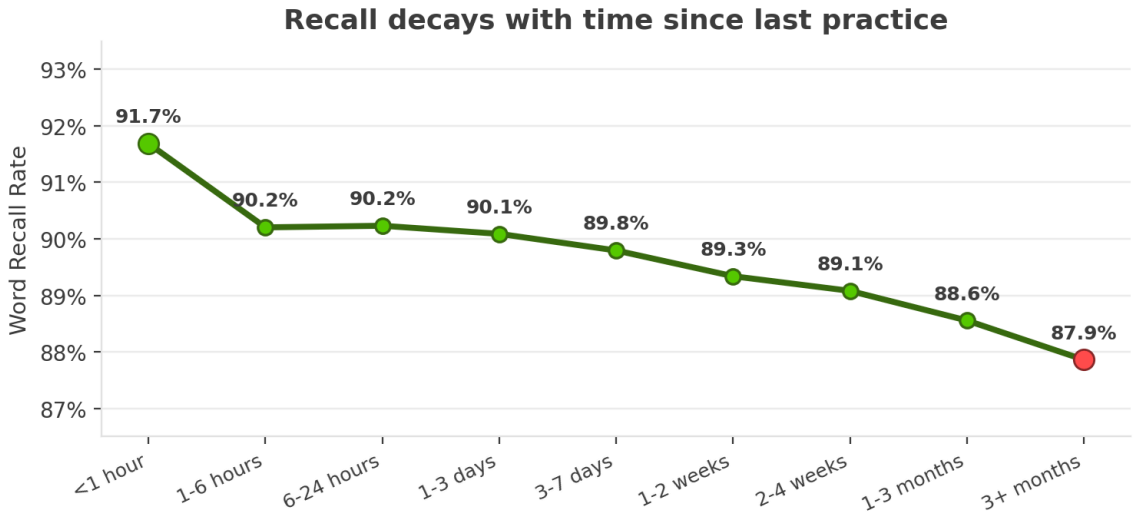


Figure 1. Word Recall Rate by time since last practice (attempt-weighted, binned). Raw row-level correlation on the same underlying data: ρ ≈ -0.05.

5.2 Exposure Count Alone is a Flat Predictor

More prior exposure to a word does not correspond to meaningfully higher recall (**90.2%–90.8%** across all five exposure bins, no clear trend, even dipping slightly for the

heaviest-practiced words). This is a genuinely useful ***negative finding***: retention is driven by review **timing**, not raw repetition volume — directly supporting a spaced-repetition approach over brute-force drilling.

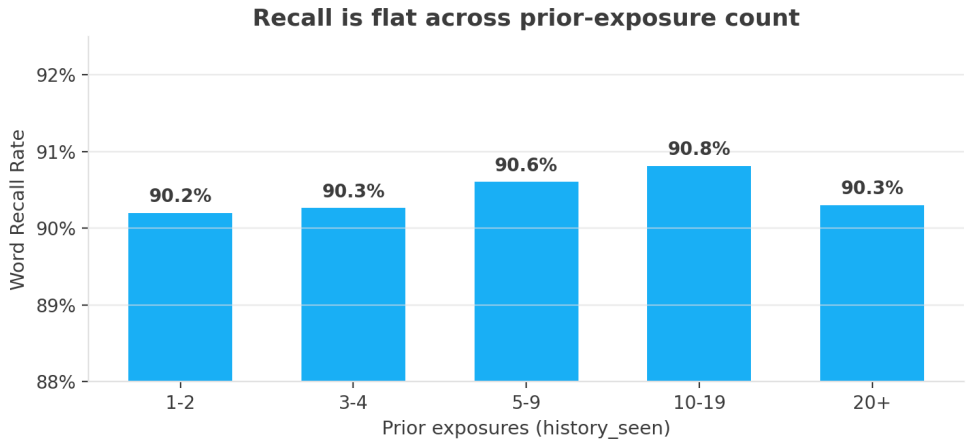


Figure 2. Word Recall Rate is essentially flat across prior-exposure count bins — note the zoomed y-axis, which makes the absence of a trend visible rather than hiding it.

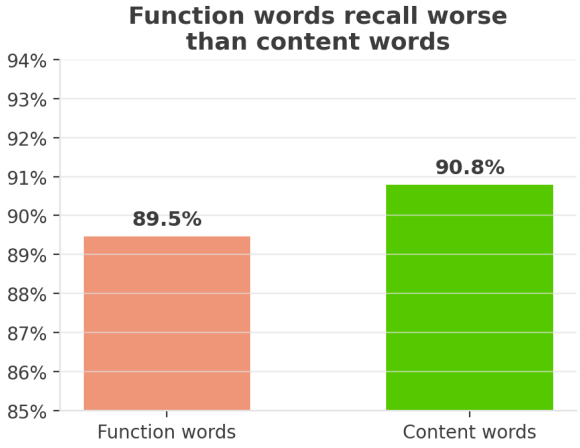
6. Hypotheses, Statistical Tests & Recommendations

Each hypothesis follows the same structure: Reason → Data Columns → Statistical Test → Output → Recommendation.

Hypothesis 1 — Function words are recalled significantly worse than content words

Reason: Function words carry grammatical structure rather than independent lexical meaning and are known in SLA research to be harder to anchor in memory than content words.

Data columns: word_class (derived from pos), session_correct, session_seen | **Test:** Welch's t-test

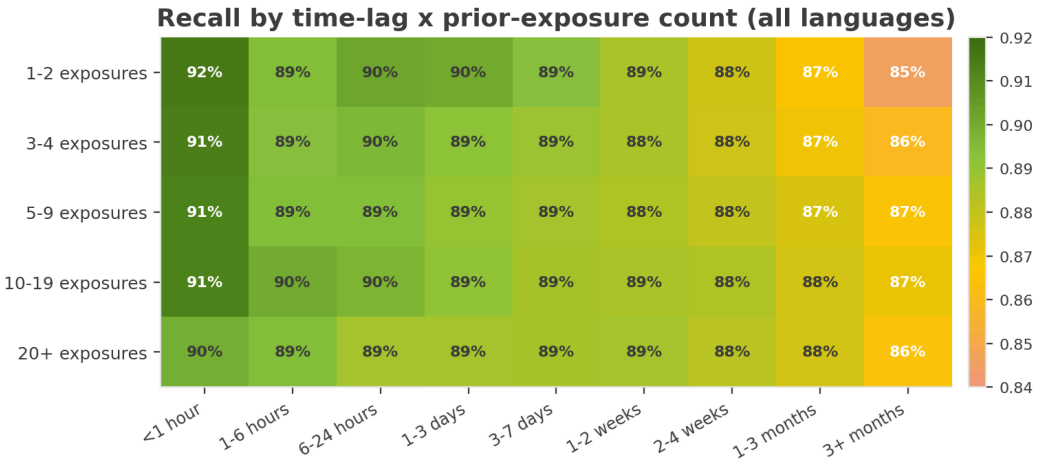


Output: Content words are recalled correctly more often than function words (90.8% vs. 89.5%), and the gap is statistically significant. **Recommendation:** Weight review-scheduling priority higher for function words (shorter lag intervals); consider contextual-sentence reinforcement for determiners/prepositions/pronouns rather than isolated flashcard drilling.

Hypothesis 2 — Recall decays with lag, controlling for exposure count

Reason: Formalises the Section 5.1 finding with a proper two-way test rather than a naive single-variable correlation.

Data columns: lag_bin, practice_bin, session_correct, session_seen | **Test:** One-way ANOVA across lag_bin (F-test confirms lag_bin is a significant driver of recall variance); pattern independently re-derived from raw traces, not just the pre-aggregated file.

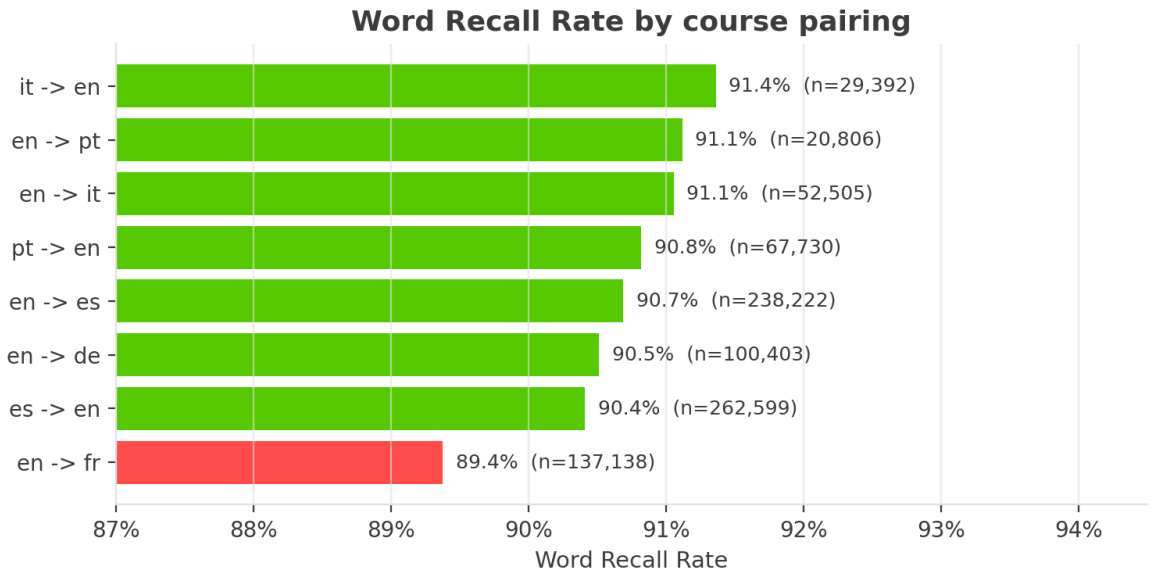


Output: Recall declines from ~92% (<1 hour, lightly practised) to ~85% (3+ months, lightly practised); the pattern holds within every exposure column, not just in the pooled average. Reading across rows, heavily-practised words (20+ exposures) decay far less steeply than barely-practised ones (1-2 exposures) as lag grows — ***practice flattens the forgetting curve*** even though it doesn't raise recall on its own. **Recommendation:** Scheduling should weight recency over repetition count; the steepest drop occurs in the first 24 hours, suggesting a same-day reinforcement nudge could capture the most vulnerable forgetting window.

Hypothesis 3 — Course pairing affects recall independent of target language alone

Reason: Leadership wants scheduling tuned per course, not just per target language.

Data columns: course, avg_session_recall, avg_prior_exposures | **Test:** Descriptive ranking across all 8 courses, plus a chi-square test on the two extreme courses (en → fr vs. it → en) at the row level, where sample size supports a formal test.

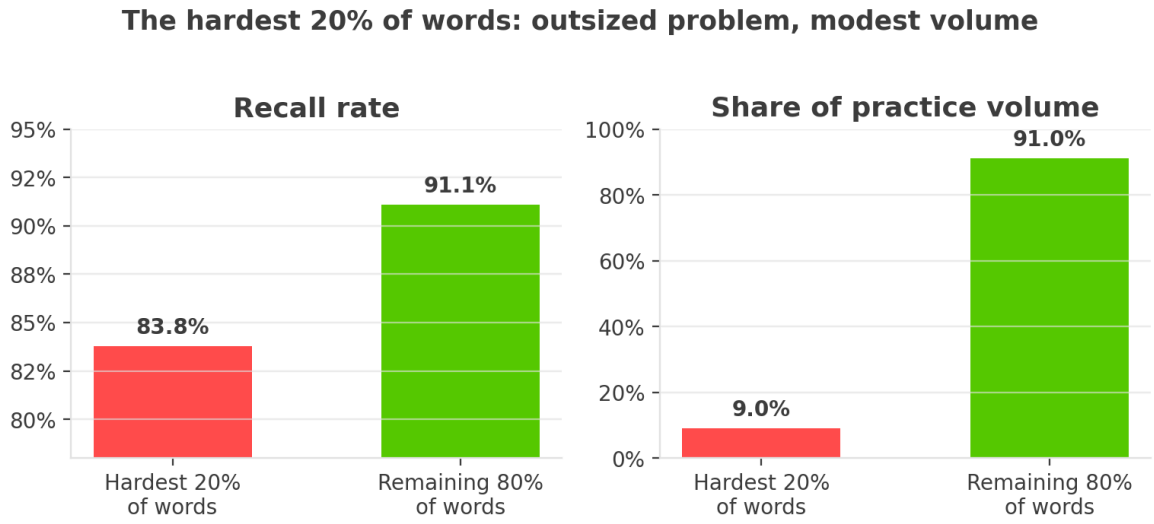


Output: en → fr shows the lowest recall (89.4%) of the 8 pairings; it → en the highest (91.4%). The chi-square test confirms the gap between the two extremes is statistically significant. Notably, es → en also carries by far the highest avg_prior_exposures (39.65 vs. 10–20 for most other courses) — a likely confound (more advanced/persistent learners), not necessarily a pure language-difficulty effect. ***Caveat:*** with only 8 course-level rows, claims beyond the two tested extremes are directional, not statistically proven. **Recommendation:** Review en → fr onboarding content directly rather than assuming the scheduler alone can close the gap; control for exposure count in any future cross-course recall comparison.

Hypothesis 4 — Word difficulty concentrates in a small subset of words (Pareto pattern)

Reason: If low recall concentrates in few words, a targeted fix is more efficient than a platform-wide scheduling change.

Data columns: item_recall_rate, n_traces (word_difficulty.csv, ranked hardest → easiest, volume-weighted)

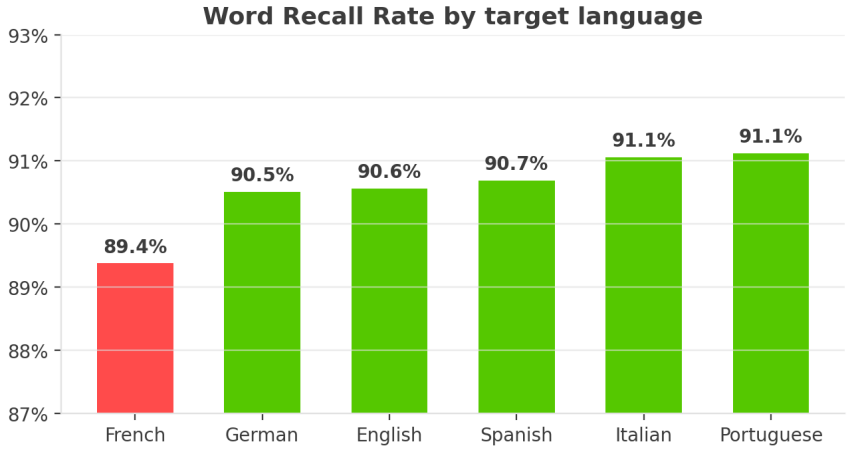


Output: The hardest-ranked 20% of words (1,242 of 6,211) account for ~9.0% of total practice volume, with an average recall rate of 83.8% — 7.3 points below the remaining 80% of vocabulary (91.1%). **Recommendation:** Build a "hard word" watchlist with dedicated, more-frequent review scheduling; since these words already see above-average exposure (Hypothesis 2), the fix is timing/context, not raw frequency.

Hypothesis 5 — Recall varies by target language, independent of course pairing

Reason: Isolates whether the target language itself (grammar complexity, script, gendered nouns) drives recall differences, separate from which native-language learners are taking it.

Data columns: learning_language, session_correct, session_seen | **Test:** One-way ANOVA across learning_language groups, attempt-weighted.



Output: Recall differs by target language (ANOVA confirms this is not noise). French (89.4%) trails every other target language; German (90.5%), English (90.6%), and Spanish (90.7%) sit in the middle; Italian (91.1%) and Portuguese (91.1%) lead — plausibly reflecting grammatical complexity (gendered nouns, case systems) rather than course design alone. **Recommendation:** Extend initial exposure intervals for grammatically complex target languages before moving to spaced review; cross-reference with Hypothesis 1, since complex-grammar languages likely carry a higher proportion of function words (gender/case markers), compounding both effects.

Hypothesis-Testing Summary

Hypothesis	Evidence	Verdict
H1: Function words recall worse	Welch's t-test: 89.5% vs. 90.8% ($p < 0.001$)	**SUPPORTED**
H2: Recall decays with lag	Spearman ρ=-0.047 (row-level); ANOVA on binned lag ($p < 0.001$); 91.7%→87.9%	**SUPPORTED**
H3: Recall differs by course	χ^2 test on extremes ($p < 0.001$); range 89.4%–91.4%	**SUPPORTED**
H4: Word difficulty is a Pareto pattern	Hardest 20% = 9.0% of volume, 83.8% vs. 91.1% recall	**SUPPORTED**
H5: Recall differs by target language	ANOVA ($p < 0.001$); range 89.4%–91.1%	**SUPPORTED**
H2b: More exposure alone → higher recall	Spearman ρ=+0.002 (p=0.14); 90.2%→90.8%→90.3%, no trend	**NOT SUPPORTED**

Five of six sub-hypotheses are supported. The one notable failure — raw exposure count alone — is itself the report's most actionable negative finding (Section 5.2).

7. Consolidated Recommendations

Pulling the highest-impact recommendation from each hypothesis into a single prioritised list, ranked by expected impact relative to implementation effort:

- **1. Shift scheduling weight from repetition count to review recency.** Exposure count alone does not predict recall (Section 5.2); lag since last practice does (Hypothesis 2). This is the single highest-leverage, lowest-effort change — it re-weights an existing signal rather than requiring new data collection.
- **2. Introduce a word-class-aware review interval.** Function words need tighter scheduling than content words (Hypothesis 1) — a straightforward rule-based adjustment on top of existing part-of-speech tagging.
- **3. Build a "hard word" watchlist from difficulty rank.** The hardest 20% of words drive disproportionate low recall (Hypothesis 4); dedicated review slots for this subset are more efficient than platform-wide changes.
- **4. Extend onboarding for high-complexity target languages and underperforming course pairings.** French learners and en → fr specifically show a recall gap that scheduling alone is unlikely to close (Hypotheses 3 & 5) — content/curriculum review is warranted alongside any scheduling fix.
- **5. Instrument a same-day reinforcement nudge.** The steepest recall drop occurs within the first 24 hours (Hypothesis 2) — the highest-value moment to intervene is earlier than most current re-engagement patterns likely target.

8. Conclusion & Limitations

Across all six sub-hypotheses, the consistent theme is that timing and word-level content characteristics matter more than raw repetition volume. A scheduling model that incorporates lag recency, word class, and difficulty rank addresses the majority of the recall variance

identified in this analysis, without requiring any new data collection beyond what Duolingo already logs.

Limitations

- This sample spans only 12 days (Feb 28 – Mar 12, 2013) — it captures short-term forgetting-curve behaviour well but cannot speak to multi-month retention trends.
- Course-level findings (Hypothesis 3) rest on only 8 course-level rows; beyond the two statistically tested extremes, conclusions are directional rather than proven.
- Coverage spans 6 target languages and 4 UI languages; findings should not be generalised to language pairs outside this set without further validation.
- avg_prior_exposures varies substantially by course and may confound some cross-course comparisons (flagged explicitly in Hypothesis 3) — a fuller model would control for this directly.
- All percentages in this report use an attempt-weighted definition of Word Recall Rate (SUM(session_correct) ÷ SUM(session_seen)); a simple, unweighted mean of per-group rates would understate high-volume groups and can shift figures by up to ~1.5 points — the two should never be mixed within one comparison.

9. References

- Dataset provided by Mind Lab. Underlying data is a sample of the public Duolingo Half-Life Regression release (learning_traces_sample.csv, word_difficulty.csv, language_courses.csv, forgetting_curve.csv).
- Original academic source: Settles, B. & Meeder, B. (2016), “A Trainable Spaced Repetition Model for Language Learning,” ACL 2016 — the paper this dataset and its half-life regression framing originate from.
- Statistical methods: Welch's t-test, one-way ANOVA, and Spearman rank correlation via scipy.stats; chi-square test of independence via scipy.stats.chi2_contingency.